

Generating 4D Robot-Object Interactions by Distilling Video Diffusion Models

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Abstract—Explicit 4D interaction data capture evolving 3D geometry and spatial relationships, supporting learning and planning for robotic manipulation. However, large-scale 4D datasets capturing coordinated robot motion and scene deformation remain scarce. We address this gap by distilling a video diffusion model into a hybrid 4D Gaussian Splatting representation. Mesh-anchored scene Gaussians capture deformation, while link-attached robot Gaussians preserve the robot’s kinematic structure. Our two-stage optimization first jointly generates scene and robot motion through video score distillation with geometric and temporal regularization, then freezes the scene and refines the robot trajectory using contact objectives. Across ten manipulation tasks, our method achieves higher cross-view feature consistency and stronger semantic alignment than the evaluated 4D generation baselines, while geometric and kinematic evaluation shows reduced robot–object penetration, smaller contact timing errors, and smoother motion compared with a state-of-the-art 4D generation method. Ablation studies further demonstrate the contributions of the proposed hybrid 4D representation and trajectory refinement to preserving geometry and improving contact consistency. Human evaluations also show greater visual interaction plausibility and robot consistency than the evaluated video models. These results support the generation of structured 4D robot–object interaction data from video priors without requiring a large-scale robot-specific 4D training dataset, with demonstrated applications in downstream robot policy learning. Project Page: <https://anonymousprojectpage.github.io/predict4dgsroi/>

I. INTRODUCTION

Robotic manipulation takes place in a 4D world, where three-dimensional geometry and spatial relationships evolve over time [1], [2]. To interact effectively with its environment, a robot must account for where objects are, how they move and deform, and how its own motion establishes and maintains contact with them. Explicit 4D interaction data makes these quantities accessible within a shared spatial and temporal representation. It can support geometrically consistent observations across viewpoints, provide motion supervision for policy learning, and enable the assessment of spatial constraints during trajectory planning. Generating such data therefore offers a promising foundation for robot learning and manipulation.

Despite its value, acquiring diverse 4D robot–object interaction data is difficult [3], [4]. High-quality capture typically requires many synchronized cameras with accurate intrinsic and extrinsic calibration. During manipulation, the robot and object frequently occlude each other, leaving contact regions and other surfaces unobserved. For robotics, this information must additionally be aligned with articulated robot configurations and robot–object interactions. This scarcity constrains the direct training of 4D generative models. Existing feed-

forward methods can generate individual object motion [5], but extending these capabilities to robot–object interactions is constrained by limited training data.

Pre-trained video generative models offer a valuable source of motion knowledge for addressing this data gap. Learned from large video collections, these models encode priors about how objects move, deform, and interact. Distillation provides a way to transfer these priors into explicit 4D representations: a candidate 3D motion sequence is differentially rendered, and the video model supplies guidance to optimize its underlying spatial and temporal parameters. Several methods distill pre-trained diffusion priors into explicit 4D representations [6], [7]. However, these methods can only generate for single-object animation. CHORD extends video-prior distillation to multi-object interactions [8]. However, the generated robot–object interactions may exhibit inconsistent contact or robot poses that cannot be realized within the robot’s joint limits. To address these issues, we propose a hybrid 4D Gaussian Splatting framework combining mesh-anchored scene Gaussians with link-attached robot Gaussians. Joint-position bounds constrain robot motion, while contact objectives guide trajectory refinement to improve contact consistency.

Our framework generates 4D interactions through two-stage optimization. In the first stage, we distill a pre-trained video diffusion model through score distillation sampling, optimizing scene deformation and robot motion under mesh-based geometric regularization. In the second stage, we hold the generated scene motion fixed and refine the robot’s joint trajectory through differentiable optimization with contact objectives. This process converts video motion priors into explicit scene motion and robot joint configurations. The resulting interaction sequences provide structured 4D data that can be rendered from different viewpoints, while the separate robot and scene representations support adaptation across robot embodiments and visual backgrounds without retraining the video prior.

Our method achieves stronger semantic alignment and cross-view feature consistency than the evaluated 4D generation baselines. Compared with CHORD, it reduces mean robot–object penetration and contact timing error by factors of 14.4 and 5.4, respectively, while yielding smoother robot motion. Human evaluations also show greater visual interaction plausibility and robot consistency than the evaluated video models. We further conduct ablation studies to assess the contributions of our hybrid representation and trajectory refinement. When mesh anchoring and robot link attachment are removed separately, neither variant produces valid ge-



Fig. 1. We generate 4D robot-object interactions by distilling video models into Gaussian Splatting representations. Differentiable trajectory optimization adapts robot motion to the generated scene and to different embodiments. The generated data can also be used for real-world policy learning.

ometry and trajectories in any of the ten evaluated cases. Trajectory refinement reduces mean penetration from 12.37 to 1.27 mm and mean joint jerk by 43.6% relative to the unrefined trajectories. We demonstrate a downstream application by using the generated 4D interactions to generate training data for robot policy learning.

Our contributions are threefold:

- We propose a framework that generates 4D Gaussian Splatting representations of robot-object interactions without requiring large-scale 4D training datasets.
- We develop a hybrid 4D representation and two-stage optimization to improve contact quality and motion smoothness in generated robot-object interactions.
- Our method outperforms CHORD on interaction quality and evaluated 4D baselines on semantic alignment and cross-view consistency, while enabling policy-training data generation as a downstream application.

II. RELATED WORKS

4D Generation. Recent feed-forward methods such as AnimateAnyMesh [5] enable efficient 4D generation by learning motion distributions from dynamic 3D data. However, their coverage and generalization depend on the availability and diversity of 4D training data. Optimization-based approaches instead exploit pretrained image and video priors [6], [7], [9]. For example, 4D-fy [7] combines multi-view image

and video diffusion priors through hybrid score distillation. CAT4D and Animate3D further use generated multi-view videos to guide 4D reconstruction and animation [10], [11]. CHORD [8] generates multi-object interactions, but its deformation-based representation cannot itself enforce robot joint limits or rigid-link structure. Our hybrid 4DGS representation addresses these limitations by jointly distilling mesh-anchored object deformation and robot joint trajectories under URDF kinematics and joint-position bounds. Subsequent trajectory refinement aligns robot contacts with the evolving object surface, reducing penetration and improving contact consistency and motion smoothness.

4D Representations. 3D Gaussian Splatting (3DGS) enables real-time, high-fidelity rendering [12], and has been extended to dynamic 4D scenes [13], [14], [15]. However, modeling motion for a large number of Gaussians remains computationally expensive. SplineGS [16] uses adaptive cubic splines to compactly parameterize Gaussian trajectories, while Shape of Motion [17] employs global motion bases to regularize monocular 4D reconstruction. SC-GS [18] instead drives dense Gaussian deformation using sparse control points. However, these motion representations alone cannot enforce robot kinematics or robot-object contact constraints. We introduce a hybrid 4DGS representation that efficiently models deformable objects and articulated robots.

III. METHOD

We address 4D robot–object interaction generation using a hybrid Gaussian representation and two-stage optimization. The representation couples mesh-anchored scene Gaussians with link-attached robot Gaussians to model scene deformation and articulated robot motion. In the first stage, video score distillation jointly optimizes both components under geometric and temporal regularization. In the second stage, we freeze the generated scene motion and refine the robot trajectory using contact objectives while respecting joint-position bounds. Figure 2 provides an overview.

A. Hybrid 4D Gaussian Representation

We propose a hybrid, differentiable Gaussian Splatting framework that couples a deformable, mesh-driven scene Gaussian with a joint-driven articulated robot Gaussian. We first use SuGaR [19] to reconstruct a mesh-bound 3D Gaussian Splatting (3DGS) representation of the scene. Given the reconstructed static Gaussian scene, let $\mathcal{M}_0 = (\mathcal{V}_0, \mathcal{F})$ be a triangle mesh extracted from an input static Gaussian scene. We instantiate K Gaussian primitives on each mesh face using fixed barycentric coordinates. For a scene Gaussian i attached to face $f(i)$, let $\mathcal{V}_{f(i)}$ be the three vertices of the face and b_{ij} be its barycentric weights. Its time-dependent center is $\mu_{s,i}^t = \sum_{j \in \mathcal{V}_{f(i)}} b_{ij} v_j^t$, where v_j^t is the deformed position of vertex j at frame t .

Each Gaussian stores opacity, spherical-harmonic color coefficients, two in-plane scales $(s_{i,1}, s_{i,2})$, a small positive normal thickness ε_i , and an in-plane angle ψ_i . For a deformed face f , let u_f^t be the unit direction of a consistently chosen edge and n_f^t its unit normal. The orthonormal face frame is $B_f^t = [u_f^t, n_f^t \times u_f^t, n_f^t]$. The Gaussian orientation and covariance are

$$\begin{aligned} Q_{s,i}^t &= B_{f(i)}^t R_z(\psi_i), \\ \Sigma_{s,i}^t &= Q_{s,i}^t \text{diag}(s_{i,1}^2, s_{i,2}^2, \varepsilon_i^2) (Q_{s,i}^t)^\top, \end{aligned} \quad (1)$$

where $R_z(\psi_i)$ denotes rotation about the local face normal. Thus the Gaussian centers remain on their supporting faces and their thin axes follow the deformed surface normals.

Scene motion is parameterized by a sparse deformation graph. Graph nodes $\{p_m\}_{m=1}^M$ are sampled on the rest mesh. For each vertex v_j , we precompute a neighbor set $\mathcal{N}(j)$ and normalized skinning weights w_{jm} . A time-conditioned deformation network with parameters ϕ outputs a rotation $R_m^t \in \text{SO}(3)$ and translation $d_m^t \in \mathbb{R}^3$ for each node and frame, relative to the rest configuration. Vertex motion is obtained by embedded-deformation skinning,

$$v_j^t = \sum_{m \in \mathcal{N}(j)} w_{jm} [R_m^t(v_j^0 - p_m) + p_m + d_m^t]. \quad (2)$$

for $t = 1, \dots, T$, while the initial mesh at $t = 0$ is fixed. Thus ϕ determines the vertex sequence $\{v_j^t\}$ and the scene Gaussian state $\mathcal{G}_{\text{scene}}^t$, while the mesh topology and Gaussian-to-face bindings remain fixed.

The robot is represented by Gaussian splats reconstructed at a known reference joint configuration q_{ref} . To associate

the reconstructed Gaussians with the robot’s kinematic structure, we sample a point cloud from the URDF-defined link geometries and transform it into a common coordinate frame using forward kinematics at q_{ref} . Each sampled point retains its corresponding link index. We register the reconstructed Gaussian centers to this URDF point cloud using iterative closest point (ICP), then use the aligned point correspondences to assign each Gaussian to its corresponding robot link. After registration, each Gaussian’s center and covariance are transformed into its attached link’s local coordinate frame using the inverse link transform at q_{ref} . These bindings and local Gaussian parameters remain fixed during motion optimization.

We optimize unconstrained joint parameters $\Theta = \{\theta_t\}_{t=0}^T$. For each frame $t = 0, \dots, T$, we map θ_t to joint angles q_t within their prescribed limits using a tanh parameterization:

$$q_t = q_{\min} + \frac{1}{2} (\tanh(\theta_t) + 1) \odot (q_{\max} - q_{\min}), \quad (3)$$

where $q_{\min}$ and $q_{\max}$ denote the joint limits, $\tanh$ is applied element-wise, and $\odot$ denotes element-wise multiplication. This parameterization enforces joint-position bounds while allowing unconstrained optimization of Θ .

For each robot link, differentiable forward kinematics computes its link-to-world rotation and translation at q_t . These transforms map the attached Gaussians’ link-local centers into world coordinates and rotate their covariances accordingly, preserving each link’s reconstructed geometry. Opacity, principal-axis scales, and spherical-harmonic color coefficients remain fixed during motion optimization.

B. Score Distillation with Physical Constraints

In the first stage, we optimize the 4D representation using Score Distillation Sampling (SDS). Prior work applies SDS to diffusion models such as DDPM and DDIM [6], [20], [21], and recent studies adapt it to flow-matching models [8], [22]. We follow this direction and distill Wan 2.2 as the video foundation model. Let $x(\phi, \Theta)$ denote the rendered video induced by deformation parameters ϕ and joint parameters Θ , let $z = E(x(\phi, \Theta))$ be the latent from the Wan encoder E , and let $\mathbf{y}$ be the text condition. The score distillation pseudo-gradient is:

$$\begin{aligned} \nabla_{\phi, \Theta} \mathcal{L}_{\text{SDS}}(\phi, \Theta; z, \mathbf{y}) \\ = \mathbb{E}_{\sigma, \epsilon} \left[w(\sigma) \left(\hat{v}(z_\sigma; \sigma, \mathbf{y}) - (\epsilon - z) \right) \cdot \frac{\partial z}{\partial(\phi, \Theta)} \right]. \end{aligned} \quad (4)$$

for a sampled noise level $\sigma \in (0, 1)$ and noise $\epsilon \sim \mathcal{N}(0, I)$, $z_\sigma = (1 - \sigma)z + \sigma\epsilon$. The Wan transformer predicts velocity fields under both conditional and unconditional contexts, denoted by v_{cond} and v_{uncond} . $\hat{v}$ combines them with classifier-free guidance (CFG) as $\hat{v} = v_{\text{uncond}} + s_{\text{cfg}}(v_{\text{cond}} - v_{\text{uncond}})$, where s_{cfg} is the guidance scale.

We further regularize scene motion. Let $\mathcal{E}$ be the set of mesh edges, $\mathcal{A}$ the set of adjacent face pairs, and n_f^t the unit normal of face f . We constrain the deformation graph’s

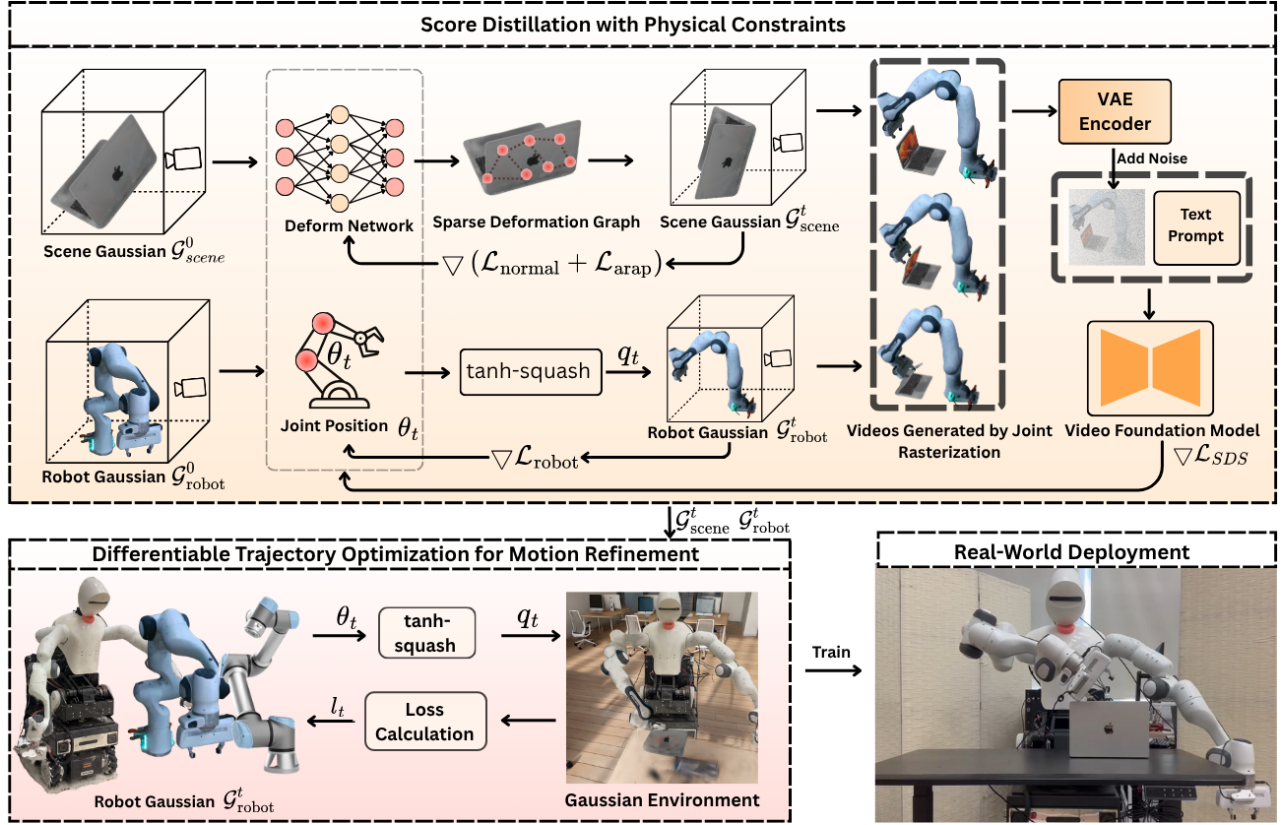


Fig. 2. **Method Overview.** Our hybrid 4D representation combines mesh-anchored scene Gaussians driven by a sparse deformation graph with link-attached robot Gaussians updated by differentiable forward kinematics. Scene and robot Gaussians are jointly rasterized and optimized by score distillation with geometric and motion regularizers, then refined by differentiable trajectory optimization.

motion using normal consistency and an edge-length preservation penalty inspired by As-Rigid-As-Possible (ARAP) regularization:

$$\mathcal{L}_{\text{normal}} = \frac{1}{T|\mathcal{A}|} \sum_{t=1}^T \sum_{(f,g) \in \mathcal{A}} (1 - (n_f^t)^\top n_g^t), \quad (5)$$

$$\mathcal{L}_{\text{ARAP}} = \frac{1}{T|\mathcal{E}|} \sum_{t=1}^T \sum_{(i,j) \in \mathcal{E}} (\|v_i^t - v_j^t\|_2 - \|v_i^0 - v_j^0\|_2)^2. \quad (6)$$

The robot trajectory is regularized by first- and second-order finite differences, together with a first-frame anchor:

$$\begin{aligned} \mathcal{L}_{\text{robot}} = & \lambda_s \sum_{t=1}^T \|q_t - q_{t-1}\|_2^2 \\ & + \lambda_a \sum_{t=2}^T \|q_t - 2q_{t-1} + q_{t-2}\|_2^2 \\ & + \lambda_{\text{anc}} \|q_0 - q_{\text{init}}\|_2^2. \end{aligned} \quad (7)$$

The complete first-stage objective is

$$\min_{\phi, \Theta} \lambda_{\text{SDS}} \mathcal{L}_{\text{SDS}} + \lambda_{\text{normal}} \mathcal{L}_{\text{normal}} + \lambda_{\text{ARAP}} \mathcal{L}_{\text{ARAP}} + \mathcal{L}_{\text{robot}}, \quad (8)$$

C. Trajectory Optimization for Motion Refinement

After score distillation, we freeze the generated scene motion and refine only the joint configurations q_t , $t = 0, \dots, T$, initialized from the SDS solution. We optimize through differentiable forward kinematics, retaining the tanh parameterization in Eq. (3) to enforce joint-position bounds.

For pushing, we select the first SDS frame whose minimum distance between candidate robot Gaussian centers and object mesh vertices is at most τ_c . The closest Gaussian defines the robot anchor; the K_c nearest vertices to its matched object vertex define a surface patch. The patch position and normal are the mean vertex position and normalized mean vertex normal. For grasping, an antipodal pair of surface anchors near the SDS gripper trajectory are paired with the finger pads. The selected contact correspondences are kept fixed during refinement. Robot-side contact positions are updated through forward kinematics, while object-side contact positions follow the precomputed mesh motion.

Let a index the contacts: one Gaussian-surface contact for pushing, or two finger-pad contacts for grasping. Denote the robot-side position, object anchor, and oriented unit surface normal by g_a^t , p_a^t , and n_a^t , respectively. With $e_a^t = g_a^t - p_a^t$, the normal and tangential distances are $d_{n,a}^t = (e_a^t)^\top n_a^t$ and $d_{\tau,a}^t = \|e_a^t - d_{n,a}^t n_a^t\|_2$. Normals initially face the robot contact and retain consistent orientation during tracking. We

use a unified contact loss:

$$\ell_{\text{contact}}^t = m_t \sum_a \left[(d_{n,a}^t - \rho_a)^2 + \lambda_{\tau,a} (d_{\tau,a}^t)^2 + \lambda_{b,a} [-d_{n,a}^t]_+^2 \right], \quad (9)$$

where $[x]_+ = \max(x, 0)$ and $m_t = \mathbf{1}[t \in \mathcal{C}]$. The interaction-frame set $\mathcal{C} \subseteq \{1, \dots, T\}$ is automatically extracted from the SDS sequence and held fixed during refinement. The offset ρ_a accounts for the desired surface separation or finger pad thickness. Contact-specific weights control tangential sliding and local penetration penalties.

We reuse $\mathcal{L}_{\text{robot}}$ from the preceding subsection, including first- and second-order joint differences and the first-frame anchor to q_{init} . The SDS trajectory supplies initialization, contact anchors, and interaction frames. The refinement objective is

$$\min_{\Theta} \lambda_c \sum_{t=1}^T \ell_{\text{contact}}^t + \lambda_{\text{robot}} \mathcal{L}_{\text{robot}}, \quad (10)$$

where Θ denotes the unconstrained joint variables in Eq. (3). We minimize this objective by first-order gradient descent.

IV. EXPERIMENTAL RESULTS

We evaluate the quality of generated 4D robot-object interactions through comparisons with state-of-the-art 4D generation methods, complemented by ablation studies on structured representations and trajectory refinement. To further examine the role of explicit spatial structure in interaction generation, we compare against video generation models. We also conduct real-world robot experiments to demonstrate the application of the generated 4D interactions to robotic manipulation.

A. Comparison with Existing 4D Generation Methods

We compare with DreamGaussian4D [6], InSpatio-World [23], SV4D [24], and CHORD [8] on the ten manipulation tasks spanning grasping, pushing, lifting, and articulated-object interaction, including grasping a banana, closing a laptop, pushing a chair, and closing a drawer. To ensure a consistent evaluation setting, all methods are provided with the same task descriptions and initial visual observations. For each method, we repeat every task ten times using different random seeds and report the results averaged over all runs. Since only our method and CHORD produce 4D Gaussian representations that permit geometric and kinematic evaluation, the corresponding metrics cannot be computed for the other baselines. We therefore evaluate all methods using CLIP Score for semantic alignment with the task description and CLIP-I for cross-view consistency. Geometric and kinematic evaluation of our method and CHORD are reported separately in Sec. IV-B.

As shown in Fig. 3, DreamGaussian4D and SV4D tend to lose fine-grained details of the robot and manipulated objects, while InSpatio-World produces a limited range of viewpoints and introduces visible artifacts. CHORD generates more

TABLE I
COMPARISON WITH 4D GENERATION METHODS.

Method	CLIP Score	CLIP-I
InSpatio-World	0.2793	0.7990
SV4D	0.2539	0.8630
DreamGaussian4D	0.2309	0.8920
CHORD	0.2907	0.9016
Ours	0.2984	0.9138

coherent 4D interactions and achieves the second-best quantitative results. However, its generation quality deteriorates over time, with the robot geometry becoming distorted and the object shape degrading in later frames. In contrast, our method better preserves the geometry and appearance of both the robot and object throughout the interaction. It also achieves the highest CLIP Score and CLIP-I in Table I, indicating better alignment with the task descriptions and stronger cross-view consistency.

B. Geometric and Kinematic Evaluation

We evaluate ten manipulation tasks using five metrics. Penetration depth measures the maximum depth of robot-object overlap. Contact timing error is $E = |t_c - t_m|$, where t_c and t_m denote first contact and object-motion onset, respectively. Contact direction score uses task-specific criteria: for pushing, $S_{\text{push}} = \hat{\mathbf{v}}^\top \hat{\mathbf{n}}$ measures alignment between unit object-motion and inward contact-normal vectors; for grasping, we use an antipodal force-closure criterion. Robot consistency measures agreement with URDF geometry, while joint jerk is $\mathbf{j}(t) = d^3 \mathbf{q}(t)/dt^3$, where $\mathbf{q}(t)$ denotes joint positions. Since CHORD [8] does not directly output joint positions, we first fit the Panda URDF model to its generated robot Gaussians and then temporally smooth the recovered joint-position trajectories. The reported CHORD results therefore evaluate the combination of its generated geometry and the fitting procedure. The fitting RMSE is $e_t = \|\mathbf{P}_t - \hat{\mathbf{P}}_t\|_F / \sqrt{N}$, where $\mathbf{P}_t$ and $\hat{\mathbf{P}}_t$ contain the N corresponding generated and fitted robot Gaussian centers. The percentage of robot-consistent frames is $C = (100\%/T) \sum_t \mathbf{1}[e_t < 20 \text{ mm}]$ over T frames.

Trajectory refinement is one of the contributions of our framework, and Table II evaluates its benefit through comparisons with the unrefined variant and ablations of its constituent losses. As shown in Table II, full refinement reduces mean penetration from 12.37 to 1.27 mm and contact timing error from 3.40 to 1.20 frames, while increasing contact-direction score from 0.579 to 0.858. Removing the contact loss degrades all three contact metrics, despite a lower joint jerk. Removing the smoothness loss retains similar contact accuracy but increases joint jerk to 19.52 rad/s³. CHORD exhibits larger penetration and contact timing errors, along with lower robot consistency. Our full refinement balances contact quality and trajectory smoothness. Importantly, our method already shows improvements over the evaluated CHORD pipeline before refinement: mean penetration decreases from 18.29 to 12.37 mm, contact timing error from

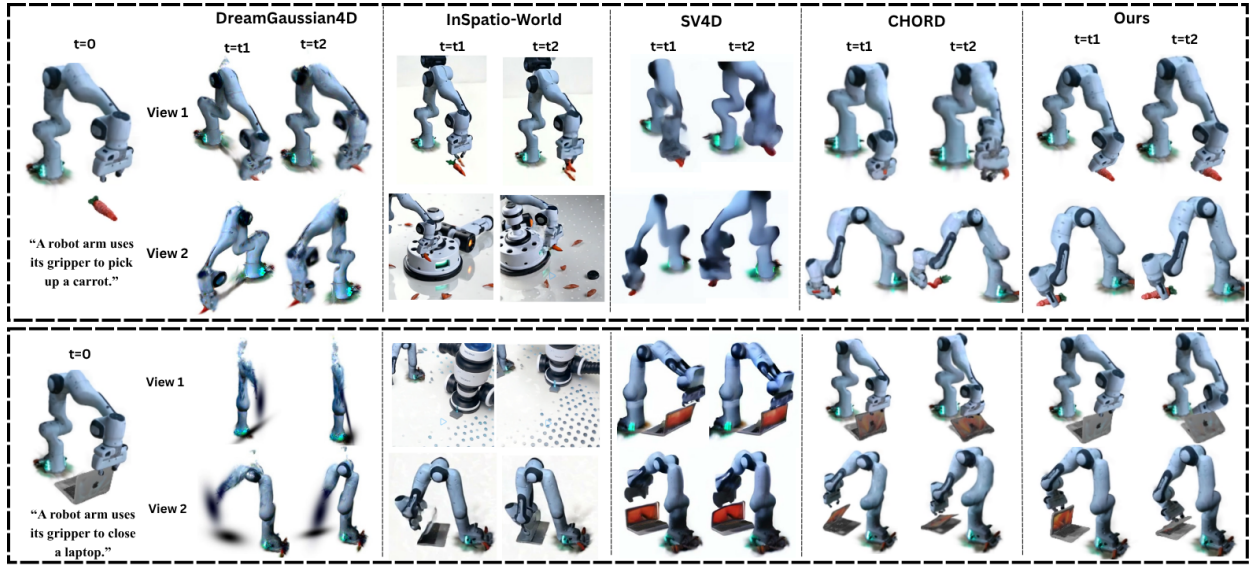


Fig. 3. **Comparison with 4D generation methods.** Baseline methods exhibit blurred details, restricted viewpoint coverage, or inconsistent robot–object geometry. Our method preserves robot morphology and maintains consistent scene geometry across views.

TABLE II

GEOMETRIC AND KINEMATIC EVALUATION ACROSS TEN MANIPULATION TASKS. RESULTS ARE REPORTED AS MEAN (STANDARD DEVIATION). BOLD INDICATES THE BEST RESULT IN EACH COLUMN.

Method	Penetration (mm) ↓	Timing Err. (frames) ↓	Contact Dir. Score. ↑	Joint Jerk (rad/s ³) ↓	Robot-Consistent Frames (%) ↑
w/o Refinement	12.37 (8.96)	3.40 (1.78)	0.579 (0.107)	9.87 (3.71)	100 (0)
w/o Smoothness	1.31 (1.81)	1.10 (0.99)	0.823 (0.136)	19.52 (7.25)	100 (0)
w/o Contact Loss	13.10 (14.57)	3.50 (1.51)	0.489 (0.203)	4.89 (2.07)	100 (0)
CHORD	18.29 (10.79)	6.50 (4.50)	0.391 (0.264)	12.35 (4.21)	32.0 (13.2)
Full Refinement	1.27 (2.13)	1.20 (1.03)	0.858 (0.120)	5.57 (2.55)	100 (0)

6.50 to 3.40 frames, and joint jerk from 12.35 to 9.87 rad/s³, while the contact-direction score increases from 0.391 to 0.579.

We remove object mesh anchoring and robot link attachment in separate ablations. Both variants achieve a 0% optimization success rate (0/10 each), where success requires valid geometry and robot trajectories suitable for geometric and kinematic evaluation. Without their respective mesh or link attachments, object and robot Gaussians drift apart. Their metrics are therefore undefined, and the variants are excluded from the quantitative comparison. These results highlight the importance of the hybrid representation for maintaining geometric and kinematic consistency in robot–object interactions.

C. The Role of Explicit 4D Structure

To examine the value of explicit 4D structure beyond monocular video generation, we compare with Wan 2.2 [25], Seedance 2.0 [26], and Veo 3.1 [27], using the same task prompts without robot-specific fine-tuning. We evaluate six viewpoints per task, yielding 60 task–view cases per model. In our user study, 26 participants visually judge *Seman-*

tic Alignment (instruction following), *Perceived Geometric Plausibility* (whether the manipulation and contact appear geometrically plausible), and *Robot Consistency* (preservation of robot appearance over time). We aggregate the individual judgments using majority voting: a case is considered to satisfy a given criterion if more than half of the participants rate it positively. We also report CLIP-T for temporal feature consistency and CLIP Score for image–text alignment.

Table III shows that our method achieves the highest rated geometric plausibility and robot consistency, together with the highest CLIP-T. Video models attain stronger semantic alignment on some metrics, but can alter robot morphology during generation (Fig. 4). The 100% robot-consistency result reflects the explicit kinematic structure of our representation. Furthermore, the robot motion can be read out directly from the 4D structure.

D. Applications

Our representation models the deforming scene and articulated robot as separate Gaussian components, allowing their assets to be composed or replaced independently. For cross-embodiment transfer, we retain the distilled 4D scene motion

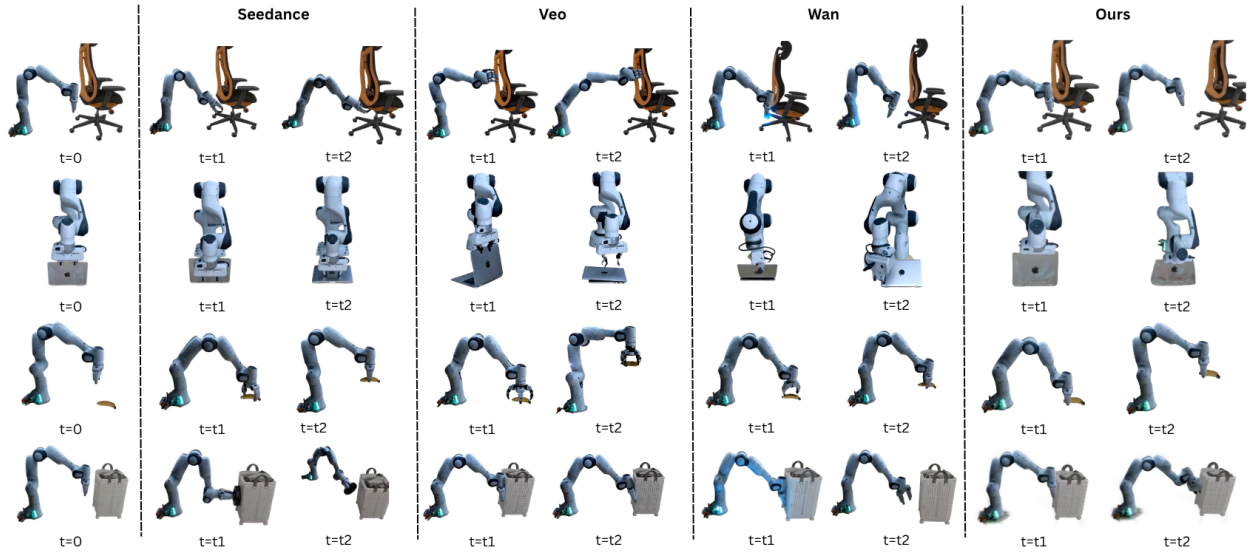


Fig. 4. **Comparisons with video models.** Existing video models often suffer from structural inconsistency, where the gripper becomes hand-like or the robot morphology changes unexpectedly.

TABLE III

QUANTITATIVE COMPARISON WITH THE VIDEO GENERATION MODELS.

Method	Sem. Align.	Geo. Plaus.	Robot Cons.	CLIP-T	CLIP Score
Wan 2.2	85.00%	63.33%	48.33%	0.9718	0.2971
Seedance 2.0	93.33%	66.67%	58.33%	0.9788	0.3050
Veo 3.1	86.67%	71.67%	28.33%	0.9694	0.3069
Ours	83.33%	81.67%	100%	0.9902	0.2984

and interaction anchors and refine only the target robot’s joint trajectory, avoiding repeated video-model distillation. We demonstrate interaction transfer to CURI, a dual-arm platform equipped with Franka arms (Fig. 6). We also evaluate transfer to a UR arm across ten manipulation tasks (Fig. 5). The UR adaptations achieve mean values of 1.89 mm for penetration depth, 2.30 frames for contact timing error, 0.794 for contact-direction score, and 5.43 rad/s³ for joint jerk, with 100% robot-consistent frames.

Video-SDS optimization takes approximately 10 hours on two A800 GPUs, while trajectory refinement, including adaptation to a different robot, takes approximately 15 minutes. Once the robot–object interaction is generated, replacing the background environment is fast, as it requires neither additional video-model distillation nor trajectory refinement. This enables efficient reuse of the same interaction across different visual environments.

As a downstream application, we train a diffusion policy using generated 4D interactions. For each task, we perform video score distillation only once and expand the resulting interaction into 800 training samples through trajectory refinement, environments replacement, and image augmentation. During trajectory refinement, we randomly sample the object’s initial position. This avoids repeated video-

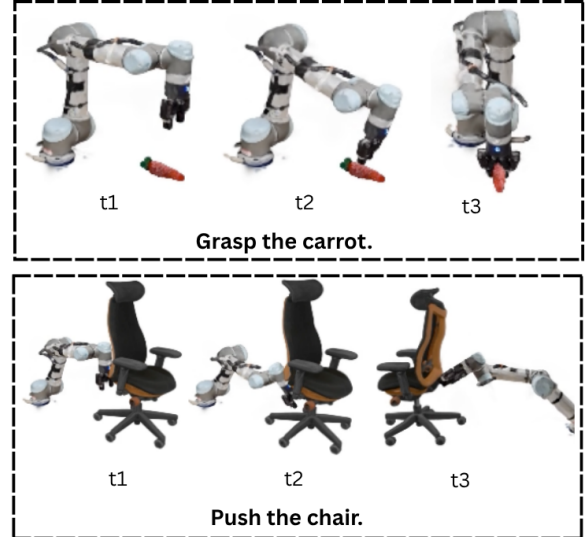


Fig. 5. **Cross-embodiment adaptation.**

model optimization, reducing the time required to expand the training dataset. In 30 real-world trials per task, the policy achieves success rates of 86.7% for closing a laptop, 86.7% for grasping a banana, 90.0% for grasping a carrot, 100.0% for closing a drawer, and 83.3% for pushing a cube. As our primary contribution lies in 4D interaction generation, we present policy learning solely to demonstrate compatibility with a downstream pipeline and therefore do not conduct baseline comparisons or ablation studies for this task.

This policy-learning example illustrates an application of our offline pipeline for generating reusable robot–object interaction data. Beyond policy learning, the explicit 4D representation supports multi-view rendering and may provide useful supervision for feed-forward 4D generators. These additional applications remain to be explored in future work.

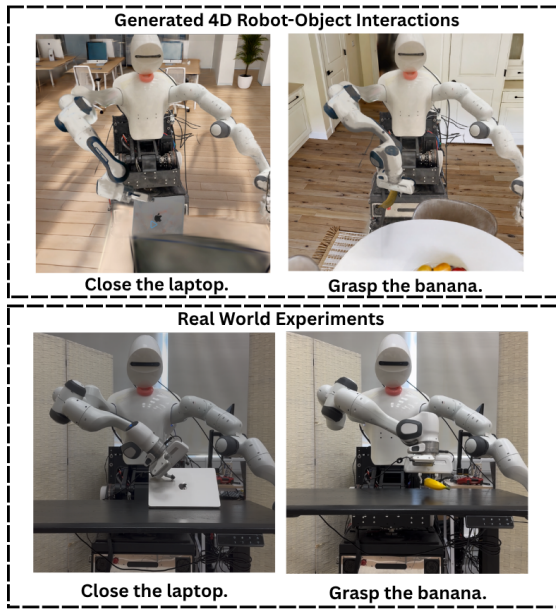


Fig. 6. Generated 4D interaction Data and Real-world deployment.

V. CONCLUSIONS

We present a framework for generating 4D robot-object interactions without large-scale 4D training data. By combining video-prior distillation with a hybrid 4DGS representation and contact-aware trajectory optimization, our method produces joint-limit-aware robot motion with consistent contact. Experiments show that our method obtains the highest average CLIP Score and CLIP-I among the 4D generation baselines, while substantially reducing penetration and contact timing errors compared with CHORD.

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